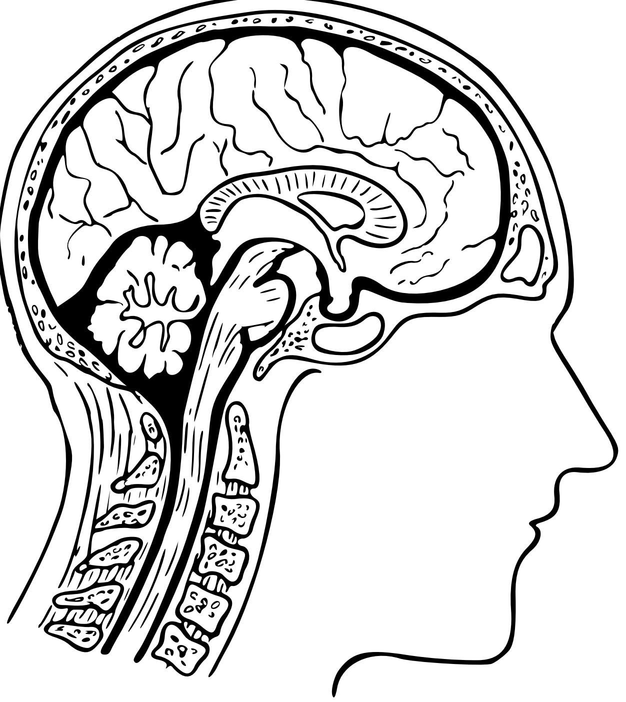
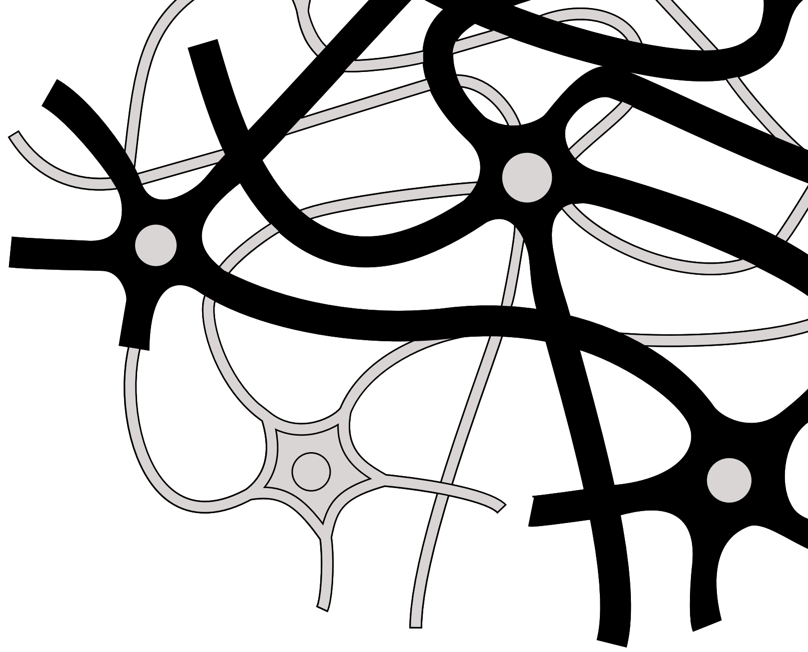
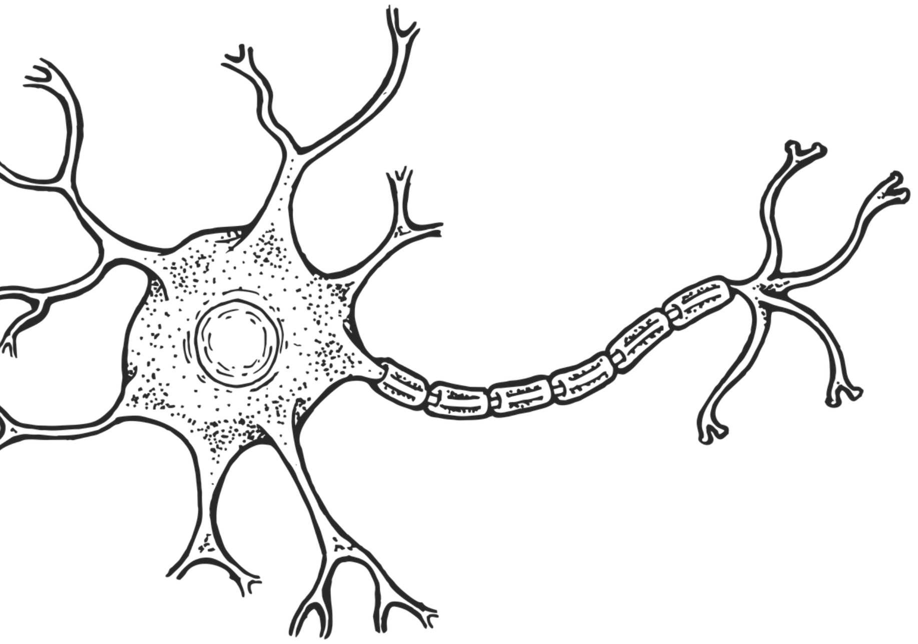
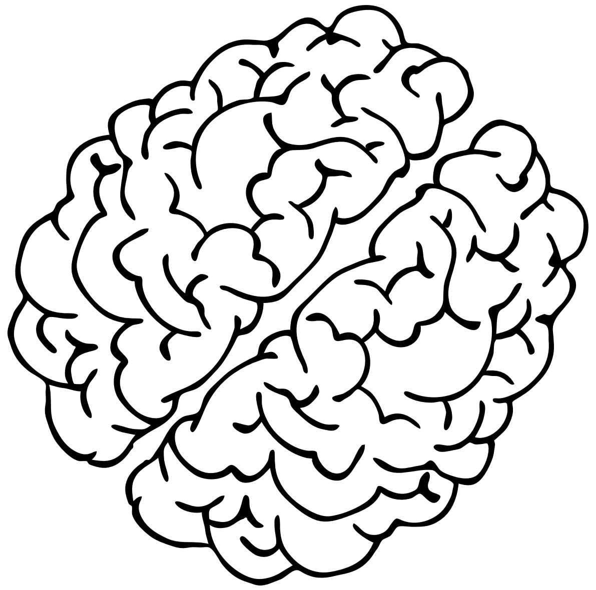


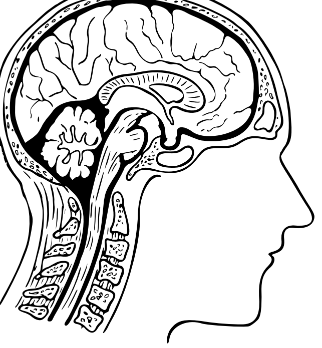


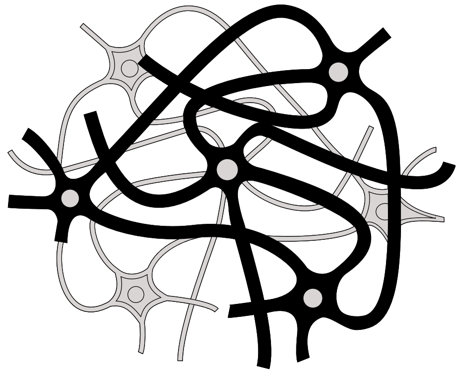
Session 10: Experimental Design for EEG

presented by Lynn









EEG Experiment Components

Paradigm Typology

Passive, Oddball, Executive, and Language designs.

Trial Structure

The anatomy of a single trial and sequence formatting.

Confounds & Controls

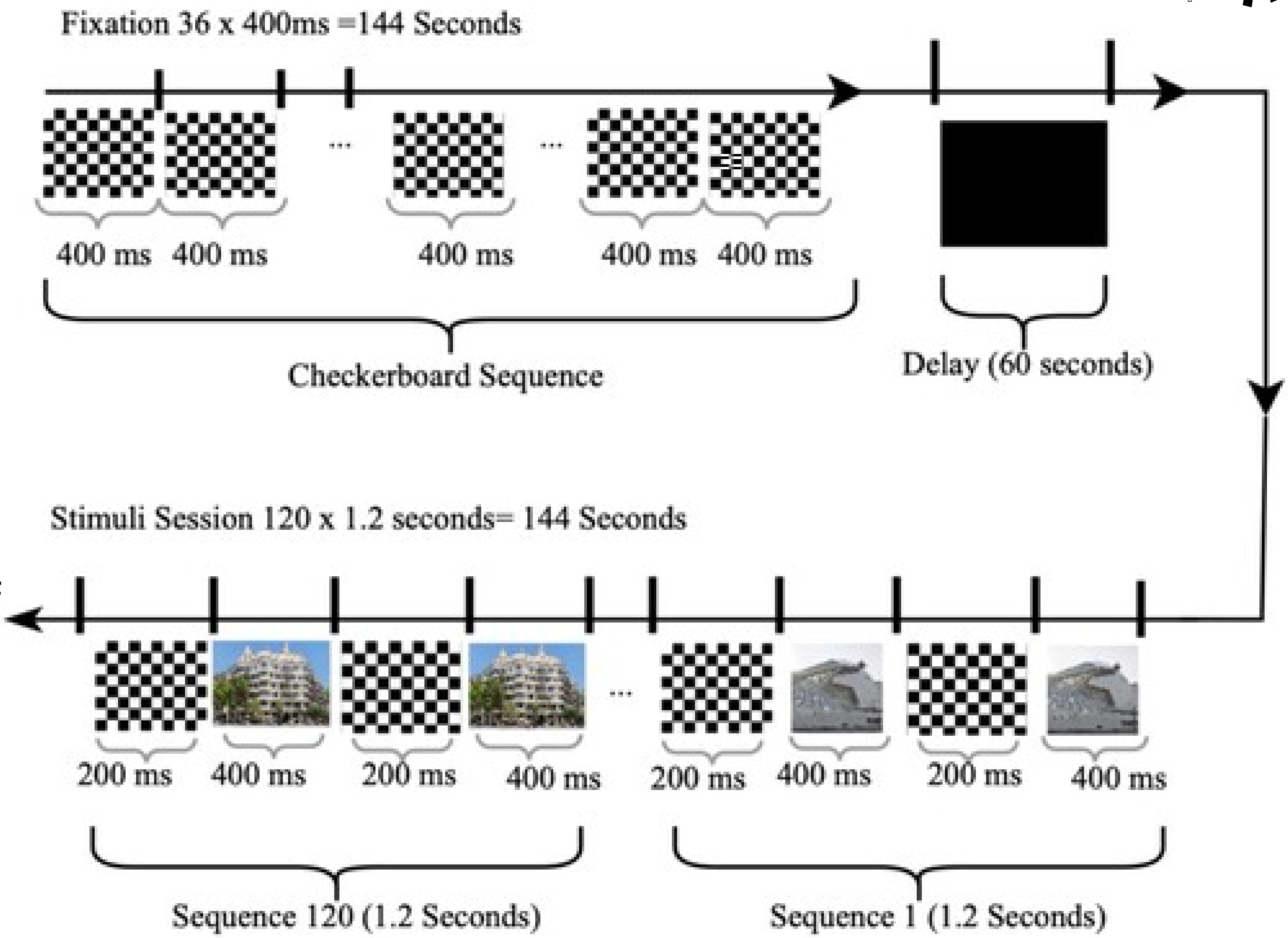
Mitigating stimulus, motor, and attention artifacts.

Group Activity

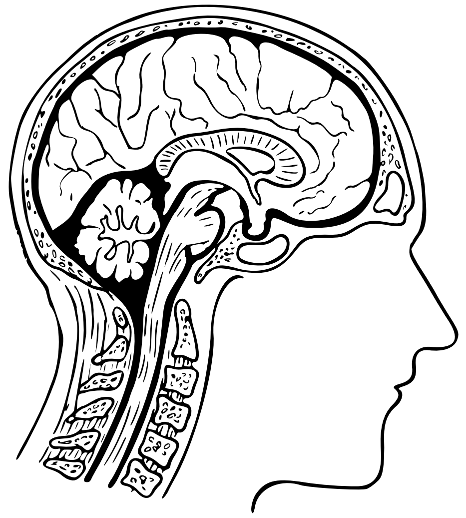
Architectural sketching of an EEG experiment.

Passive Sensory Paradigms

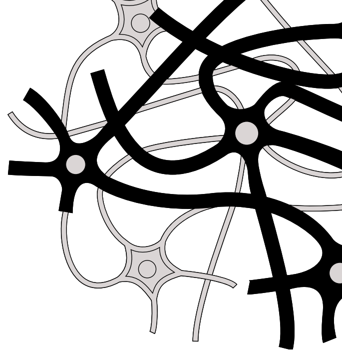
- Obligatory Sensory Responses
  - These paradigms do not require the subject to make a decision or press a button. The brain processes raw sensory input obligatorily.



- **Visual Evoked Potentials (VEP):** Checkerboard reversals or high-contrast flashes. Isolates P1, N1, and P2.
- **Auditory Evoked Potentials (AEP)**: Brief tones or broadband clicks. Isolates the auditory N1 and P2.
- **Mismatch Negativity (MMN)**: A passive deviant tone embedded in a repetitive standard sequence.



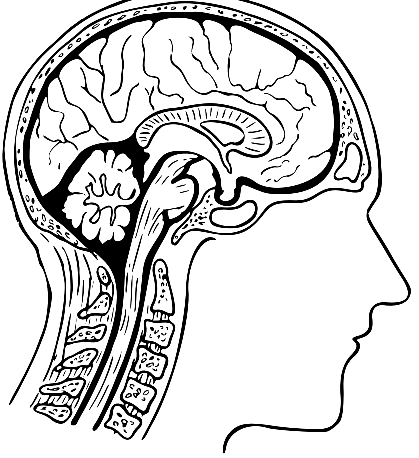
The Oddball Paradigm



Target Detection & The P300

The workhorse of cognitive electrophysiology testing active computation and context updating.

- **The Probability Matrix**: A frequent 'Standard' stimulus (80%) establishes a baseline expectation.
- **The Deviant**: A rare 'Target' stimulus (20%) violates the expectation.
- **The Task**: The subject must actively count or press a button only for the target.
- **The Readout**: Context updating generates a massive Parietal P300.



Executive Function & Memory

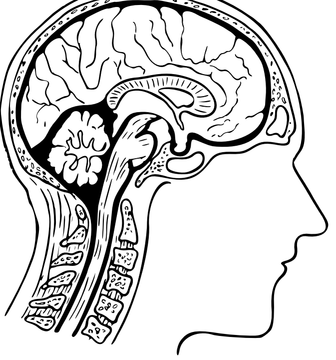
Testing the brain's regulatory mechanisms, inhibition, and maintenance.

Go/No-Go Task

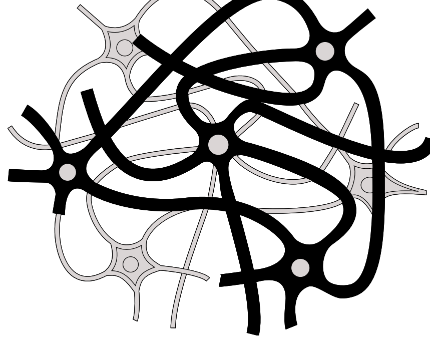
- **Respond to Stimulus A** (**Go**); withhold response to Stimulus B (No-Go).
- **The Readout**: Successful withholding generates a pronounced Frontocentral N2 and No-Go P3.

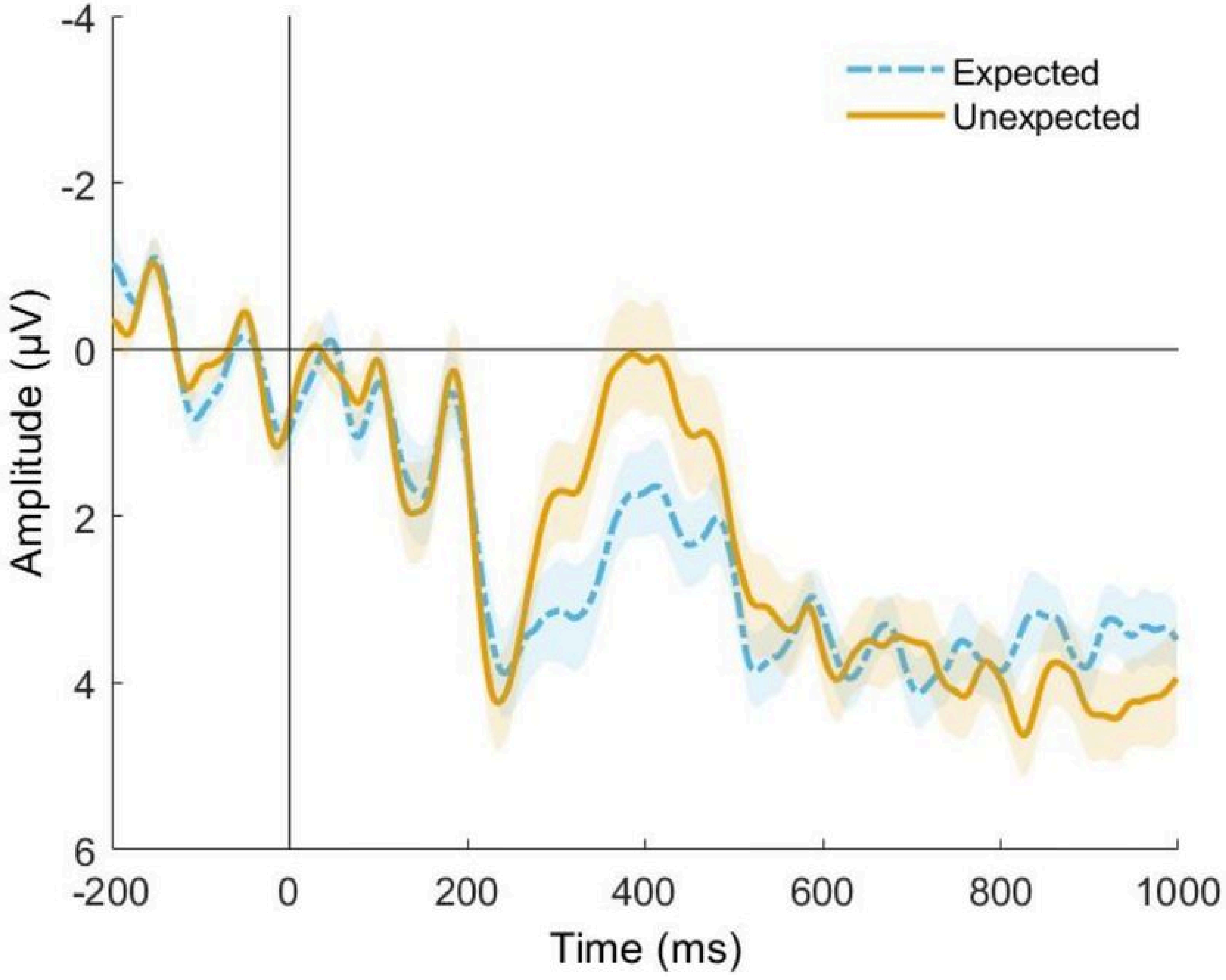
Sternberg Working Memory Task

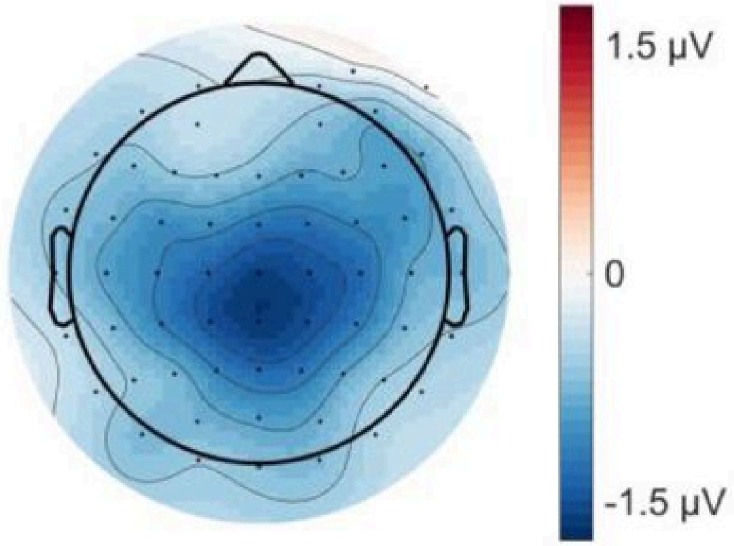
- Present a memory set (e.g., 5 letters), enforce a delay, and present a probe.
- The Readout: Sustained frontal theta power during the delay; P300 during successful recognition.



Semantic Processing







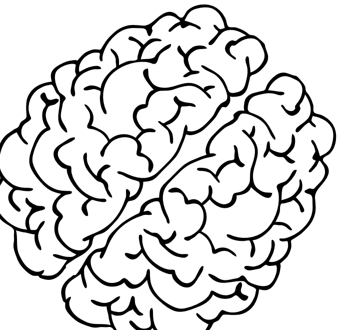
Unexpected-expected
300-500 ms

The N400 Language Paradigm

The Setup: Sentences are presented word-by-word at a fixed speed (Rapid Serial Visual Presentation).
The Contrast: The terminal word is either semantically congruent or incongruent.

The Readout

Incongruent words elicit a strong N400 over the centroparietal cortex. The amplitude scales directly with semantic incongruity.



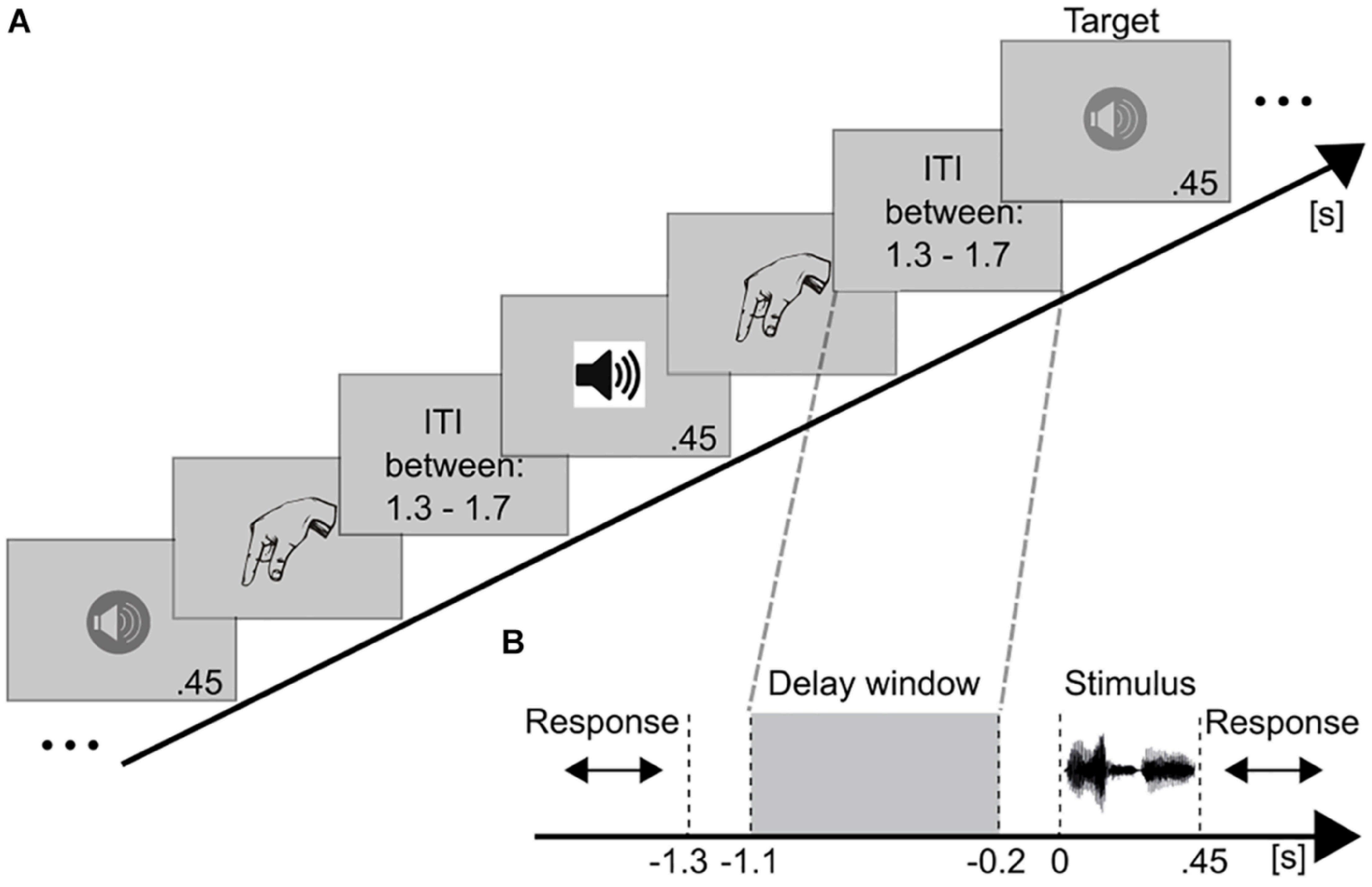
"I take my coffee with cream and sugar."

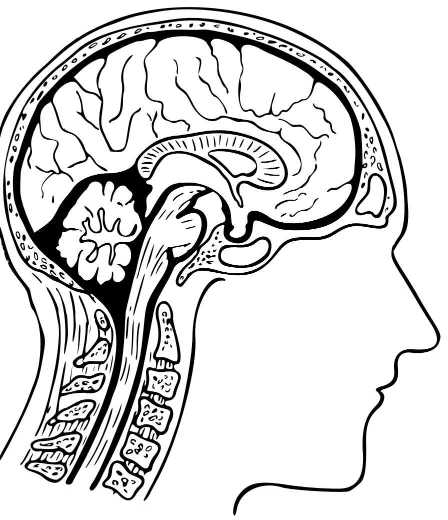
"I take my coffee with cream and dog."

Trial Structure Anatomy

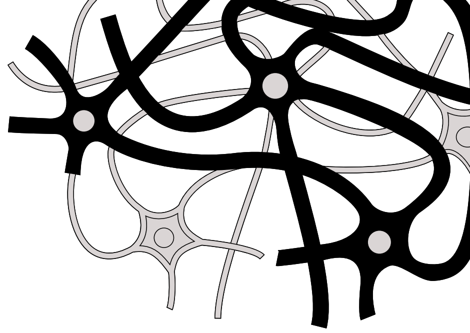
Structuring the Micro-Environment

- **Inter-Trial Interval (ITI)**: 800–1500 ms. The rest period between trials.
- **Fixation Cross**: 500–800 ms. Centers ocular focus prior to the event.
- **Stimulus Duration**: 50–300 ms (Visual) or 50–100 ms (Auditory).
- **Response Window**: 1000+ ms. Allocated time for the subject's motor execution.





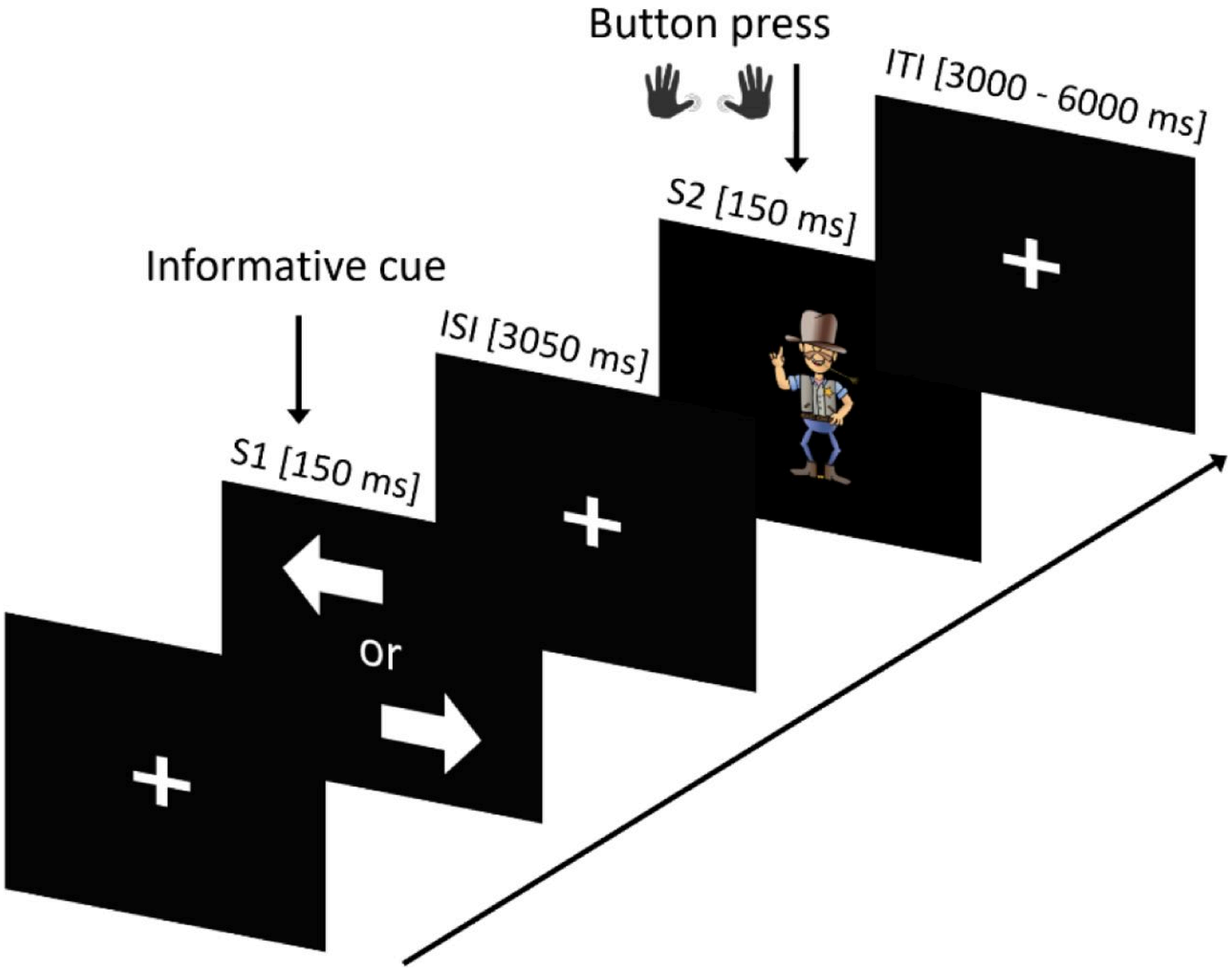
Preventing Anticipation



The Confound

Fixed timing allows the brain to predict the onset, generating preparatory slow waves (Contingent Negative Variation) that destroy the baseline.

- **Jittering the ITI:** Randomizing the interval (e.g., 800, 1100, 950 ms) is strictly required to prevent anticipation.
- **Blocked vs. Randomized:** Blocked designs maximize condition-specific focus. Randomized designs prevent expectation bias.



Confounds and Controls

Physical Confounds

Shifts in luminance, size, or auditory volume directly alter sensory ERP amplitudes.

The Control

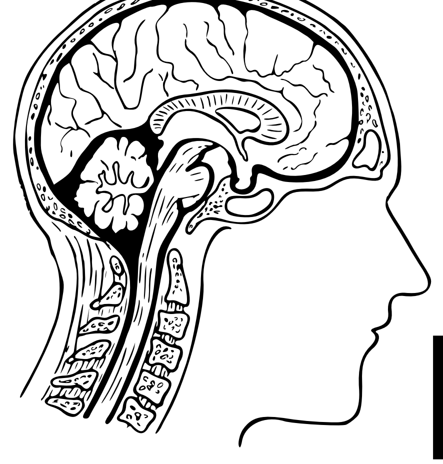
Strictly equate your stimuli for physical properties. You are testing cognition, not physics.

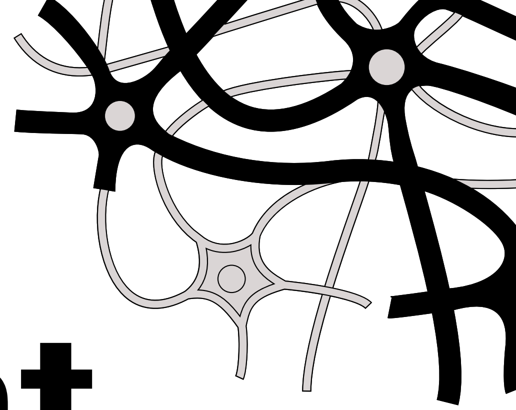
Motor Confounds

Pressing a button generates a Lateralized Readiness Potential (LRP) in the motor cortex.

The Control

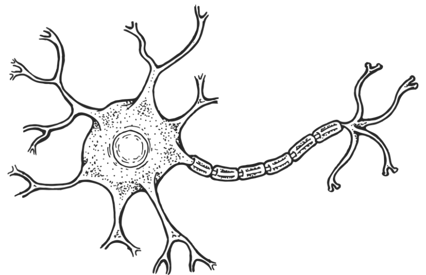
Counterbalance response hands across blocks, or delay the response prompt until after the ERP window closes.

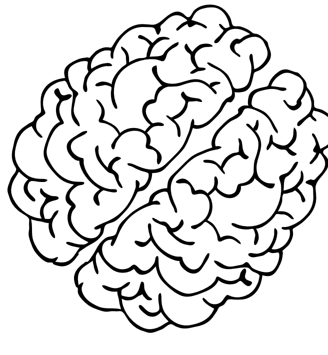


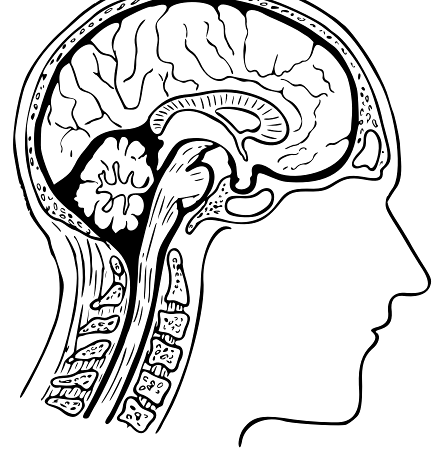


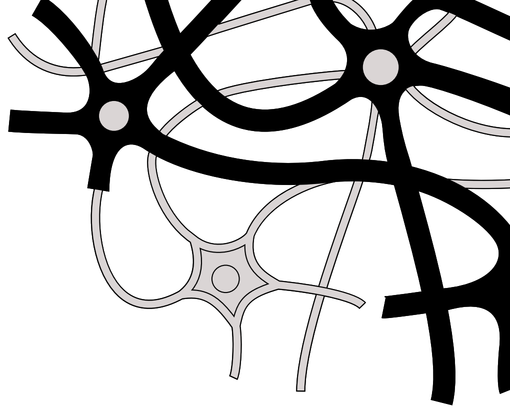
Managing the Human Element

- **Fatigue**
  - Alpha-band power increases as subjects tire, drowning out ERPs. Restrict blocks to 5–8 minutes.
- **Attention Drift**
  - Mitigated by catch trials to force continuous engagement and prevent the subject from zoning out.
- **Trial Counts**
  - A strict minimum of 30–50 artifact-free trials per condition is required to achieve a viable Signal-to-Noise Ratio.









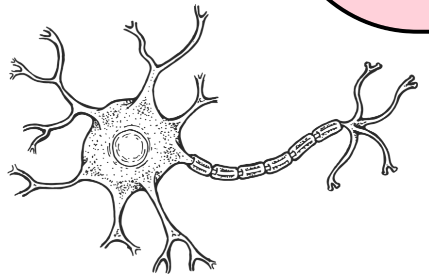
Architectural Sketch

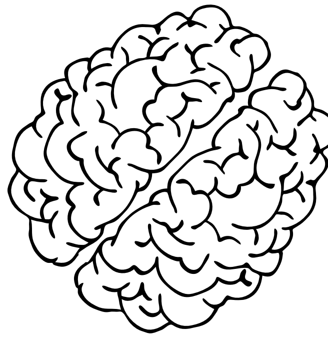
Objective

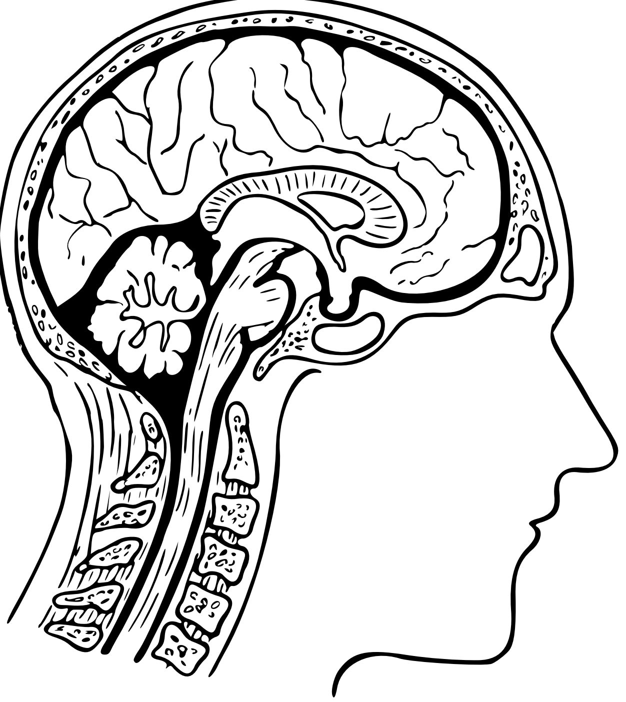
Map out the precise timing and sequence of a visual oddball task.

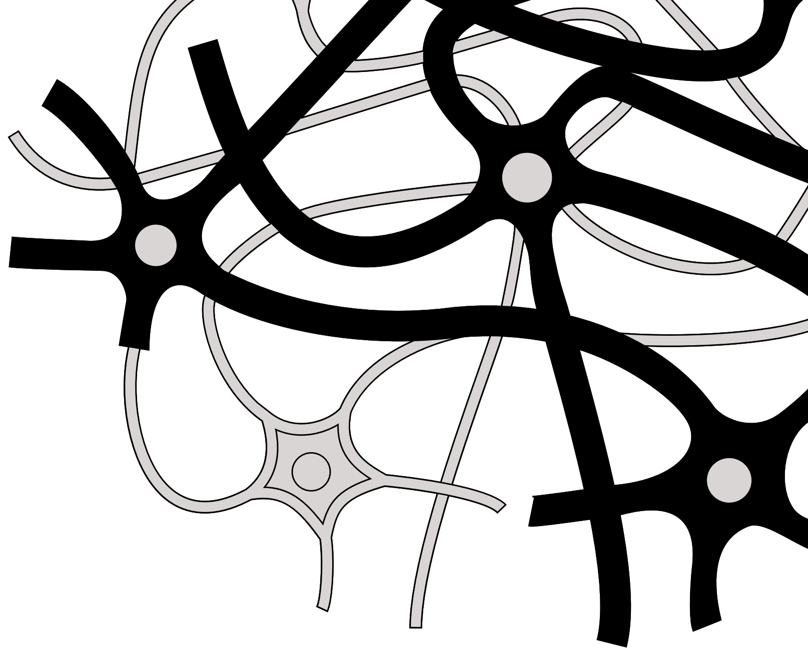
Requirements

- Define ms duration for Fixation, Stimulus, and Response.
- Define the ITI jitter range.
- Specify the total number of trials and the Target/ Standard ratio.









Thank You